Multisource Remote Sensing Interpretation of Geological Lithology via Multigranularity Adaptive Learning

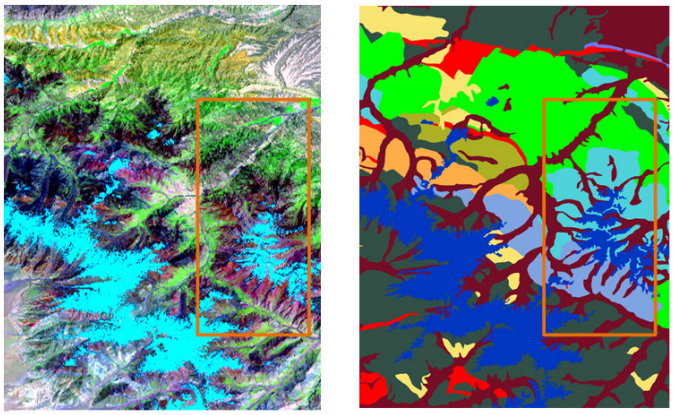
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***Abstract*—Environmental–geological interactions lead to a fragmented spatial distribution of elements and homogenization among adjacent regions. Furthermore, single-source data suffer from limited feature representation, and multisource feature fusion faces the issue of spatial heterogeneity. These issues degrade the accuracy of deep learning-based interpretation methods relative to expert visual analysis and lead to boundary ambiguity in segmentation maps. To address these issues, we propose the multigranularity adaptive learning network (MGALNet), a unified framework designed for adaptive multisource data fusion and multiple geological lithology interpretation. MGALNet enables the joint utilization of multisource data to enrich lithology details and enhance feature representation, thus addressing boundary ambiguity segmentation. Specifically, the multigranularity feature extraction and fusion module (MGFEEM) mitigates spatial heterogeneity in multisource features via double-branch granularity extraction and adaptive feature fusion, ensuring consistent feature representation. To mitigate boundary ambiguity segmentation, the feature rectify module aggregates vertical and horizontal spatial contexts to refine spatial structural features, and the spatial feature enhancement module (SFEM) captures spatial dependencies across channels to strengthen boundary feature discrimination. Experiments on the constructed Geo-WQ and Geo-XZ datasets show that MGALNet surpasses state-of-the-art methods. It achieves an overall accuracy (OA) of 83.61% and a mean intersection over union (mIoU) of 64.78% on Geo-WQ, the OA of 84.18% and mIoU of 63.99% on Geo-XZ. These results confirm that MGALNet effectively integrates multisource complementary information and achieves accurate lithological interpretation with improved computational efficiency. The code is available at https://github.com/Candy-CY/MGALNet**

***Index Terms*—Cross-source feature fusion, feature enhancement, feature rectify, geological lithology interpretation, multisource remote sensing images.**

I. Introduction

REMOTE sensing (RS) imagery, with broad spatial coverage and multidimensional observation capabilities, provides valuable information on surface lithology and stratum structures. However, geological element classification is complex, as each category (e.g., igneous and sedimentary) includes many subtypes (e.g., granite and basalt), and classification criteria based on geological age or composition often overlap [1]. Numerous geological classes challenge RS interpretation, making efficient feature extraction and accurate classification essential for automated geological resource exploration.



(a) (b)

Fig. 1. Schematic of geological elements with fragmented spatial distribution. (a) Local Landsat-8 image from the Geo-WQ region. (b) Corresponding annotated map.

Geological RS image interpretation (GRSI) relies on semantic segmentation methods and geological knowledge to extract the geological information and categorize pixels in accordance with specific geological tasks, such as lithology, soil, and glacier mapping. However, the accuracy of the current GRSI remains limited and is affected by multiple factors. Crustal movements and geological structures can result in a complex and fragmented spatial distribution of geological elements as seen in Fig. 1. This spatial form often leads to misidentification, resulting in isolated regions and blurred boundaries in the segmentation map [2]. Long-term wind and sand erosion, along with surface cover changes, have caused exposed geological elements to have similar mineral compositions, leading to serious homogenization and causing these elements to exhibit different spectra for the same object and the same spectrum for different ones [3]. The theory of regionalized variables [4] also shows that geological processes influence adjacent elements, forming spatial correlations of

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regionalized variables that can serve as important indicators for element identification [2], [5]. So, it should overall consider basic attributes (color, spectrum, and texture), local contextual information, and global interactive features in GRSI. Extracting rich and distinct detail features is essential for accurate geological interpretation. Based on the differences in feature extraction strategies, current semantic segmentation methods in GRSI can be grouped into traditional and deep learning-based approaches.

Traditional semantic segmentation methods primarily rely on visual interpretation and field investigations and require expertise in geology and RS, as well as extensive ground truth verification and data analysis, rendering these approaches labor-intensive and inefficient [6]. Although classical segmentation methods [7], [8], [9] enable rapid object segmentation, they depend on expert-defined manual parameters. Furthermore, machine learning-based segmentation methods [10], [11], [12], [13] leverage shallow image features and demonstrate satisfactory performance in simple scenarios; however, their limited generalization capability undermines their effectiveness in complex and diverse RS image segmentation tasks [14].

The success of convolutional neural networks (CNNs) in computer vision has stimulated considerable interest in their application to semantic segmentation of GRSI [15]. Unlike traditional methods, CNN automatically learn nonlinear and deeper semantic features to improve segmentation performance. The introduction of the fully convolutional network (FCN) [16] in 2015 provided a foundation for applying deep networks to RS image segmentation, and the rapid expansion of RS data has driven the development of numerous CNN-based models. Existing CNN-based methods in GRSI can be divided into single-source and multisource segmentation approaches.

Single-source semantic segmentation methods identify pixels by integrating low-level details with high-level semantics in a single image. It has been widely applied in lithology mapping [17], soil classification [18], glacier monitoring [19], and related tasks. For example, Sang et al. [20] combined the simple linear iterative clustering (SLIC) algorithm with CNN to develop an object-based SLIC-CNN method for lithological mapping using uncrewed aerial vehicle images. Liu et al. [21] designed 1-D, 2-D, and 3-D CNN models for lithology classification with hyperspectral thermal infrared data. Wang et al. [2] proposed an adversarial semisupervised segmentation network that integrates local context and global attention features in Landsat-8 images, enabling lithology mapping with limited labeled samples. However, these methods still face challenges due to the inherent traits of geological elements [22], including incomplete element recognition, blurred boundaries, and limited applicability [1]. Furthermore, the lack of auxiliary data and geological constraints restricts the ability of models to learn and capture element characteristics [23], which further affects the interpretation accuracy. Thus, integrating auxiliary data has emerged as an effective strategy to overcome the limitations of single-source data and improve interpretation accuracy across diverse scenarios.

Multisource methods fuse and utilize complementary features from different RS images, improving accuracy in complex scenarios compared with single-source methods. Han et al. [22] proposed a deep learning framework with adaptive multisource data fusion that integrates Landsat-8 and Sentinel-2 data to improve the accuracy of geological interpretation. Lu et al. [6] developed a soil element interpretation framework guided by tacit knowledge to enhance the utilization of auxiliary data and soil element interpretation accuracy. He et al. [3] proposed a sensitive feature self-aggregation deep fusion network by integrating multimodal sensitive features and prior geological knowledge, and enhanced lithology interpretation from RS images. Although multisource data can provide diverse complementary information, the spatial heterogeneity issue exists in their features. Moreover, multisource data fusion methods [24], [25] are generally designed for predefined modality combinations [3], limiting the models' applicability to other source data. Therefore, establishing an effective deep learning-based method to quickly extract distinctive features from multisource heterogeneous data and accurately identifying geological elements in diverse geological scenarios has become essential in GRSI.

The use of multisource data has become an inevitable trend in GRSI research. However, GRSI still faces several challenges, including spatial heterogeneity across different source features, limitations in generalizing multisource data fusion caused by predefined modality combinations, and boundary ambiguity segmentation resulting from fragmented spatial distribution. To resolve these issues, this study proposes the multigranularity adaptive learning network called MGALNet, a unified framework designed for adaptive multisource data fusion and multiple geological lithology interpretation. It aims to utilize multisource heterogeneous data to enrich lithological details and enhance feature representation, alleviating the problem of lithologic boundary ambiguity segmentation. The following contributions of this work are highlighted.

1) We constructed two new benchmark lithology datasets from multisource RS images, namely, Geo-WQ and Geo-XZ. Geo-WQ integrates Landsat-8 and GF-5 data, and Geo-XZ combines Landsat-8 and Sentinel-2 images because of their accessibility. These datasets are designed to evaluate the cross-source feature fusion capability and geological scenario adaptability of MGALNet and demonstrate its potential for intelligent lithology interpretation via multisource RS data.
2) To resolve the spatial heterogeneity in multisource features, we developed a multigranularity feature extraction and fusion module (MGFEEM) to adaptively learn and fuse coarse- and fine-grained features from multisource data. It can enhance the consistency of feature representation and improve lithology information utilization.
3) We proposed a feature rectification module (FRM) and a spatial feature enhancement module (SFEM) to mitigate the issue of boundary ambiguity in segmentation maps. FRM captures spatial contextual information in vertical and horizontal directions to strengthen spatial feature representation, and SFEM learns spatial dependencies

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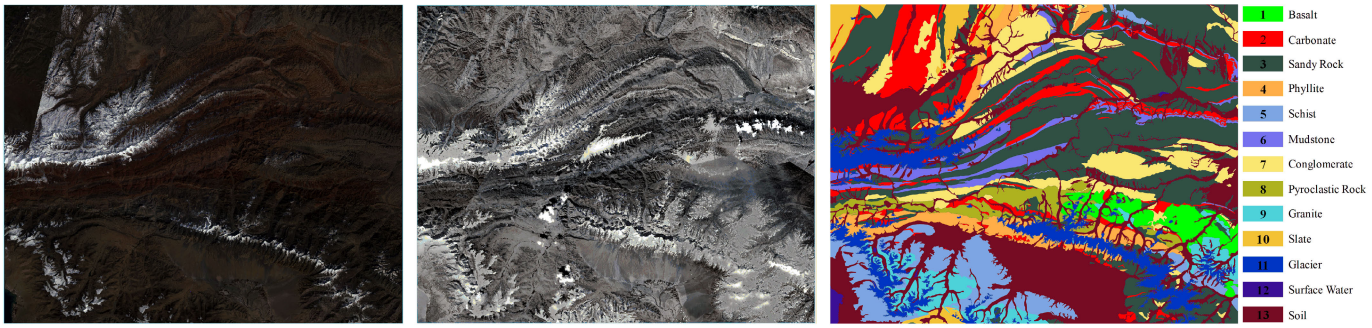


Fig. 2. Geo-WQ geological dataset comprises (a) Landsat-8 and (b) GF-5 images, and (c) label map, characterized by 13 categories and a spatial resolution of 15 m.

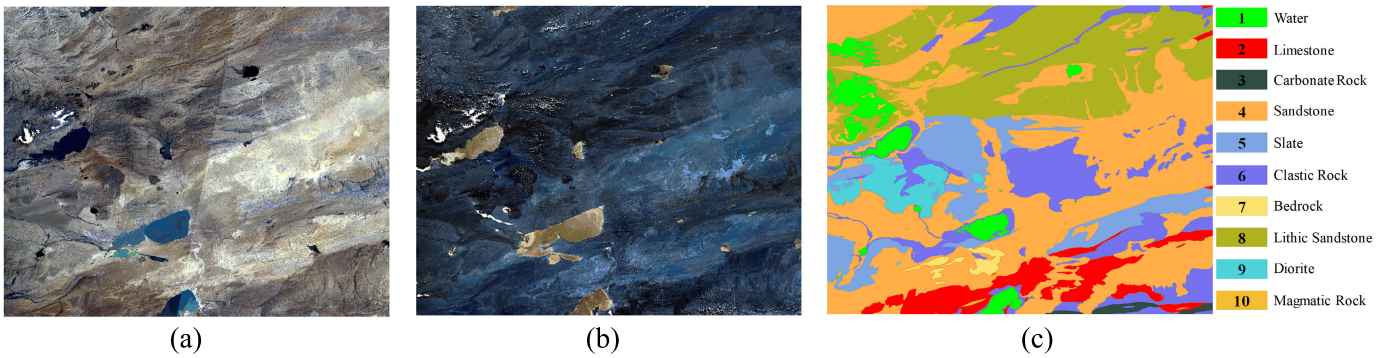


Fig. 3. Geo-XZ geological dataset comprises (a) Landsat-8 and (b) Sentinel-2 images, and (c) label map, characterized by ten categories and a spatial resolution of 15 m.

across channel dimensions to further enhance feature discrimination of adjacent elements.

This article is organized into six sections. Section I reviews related research on GRSI. Section II introduces the research area and datasets. Section III describes the MGALNet in detail. Section IV presents the results of comparison experiments, while Section V discusses the findings of the ablation study. Section VI analyzes the efficiency of models in parameters and floating-point operations (FLOPs). Section VII concludes this article and outlines directions for future research.

II. Research Area and Datasets

The research area of the Geo-WQ geological dataset is in WuQia County near Tarim Basin, where the Tianshan and Kunlun Mountains converge in Xinjiang, China. The types of rock in this area are influenced by the temperate climate, ongoing crustal movement, and erosion from external forces. Moreover, the area has experienced a complete magmatic intrusion-sedimentary process and has various lithological elements. The Geo-WQ geological dataset utilizes RS images from Landsat-8 and GF-5, both of which have a cloud coverage of less than 5%. These RS images were collected between January 1, 2023, and December 1, 2023. Fig. 2(a) and (b) displays the two preprocessed RS images of the Geo-WQ research area, and Fig. 2(c) shows their corresponding label maps. The categories of elements represented by labels 1–13 are basalt, carbonate, sandy rock, phyllite, schist, mudstone, conglomerate, pyroclastic rock, granite, slate, glacier, surface water, and soil.

The research area of the Geo-XZ geological dataset is in the northwest of Gaize County, Ngari Prefecture, Xizang, at the southern foot of Kunlun Mountains. It is adjacent to Minfeng County, Xinjiang, and is in the core area of the uninhabited Qiangtang Plateau. This region is shaped by crustal movement, volcanic activity, and plateau mountain climate, resulting in various lithological elements. This geological dataset utilizes RS images from Landsat-8 and Sentinel-2, both of which have a cloud coverage of less than 5%. These RS images were collected from January 1, 2023, to December 1, 2023. Fig. 3(a) and (b) displays the two preprocessed RS images of the Geo-XZ research area, and Fig. 3(c) presents their corresponding label maps. The categories of elements represented by labels 1–10 are water, limestone, carbonate rock, sandstone, slate, clastic rock, bedrock, lithic sandstone, diorite, and magmatic rock.

The label maps were annotated by domain experts from the Xining Comprehensive Survey Center of the China Geological Survey. The proportions of each geological element in the two datasets are shown in Fig. 4. The image sources of these two multisource geological datasets include Landsat-8, Sentinel-2, and GF-5. The following part describes the three RS image sources.

1) *Landsat-8:* The Landsat-8 platform integrates the operational land imager and the thermal infrared sensor, capturing multispectral data across 11 bands. The

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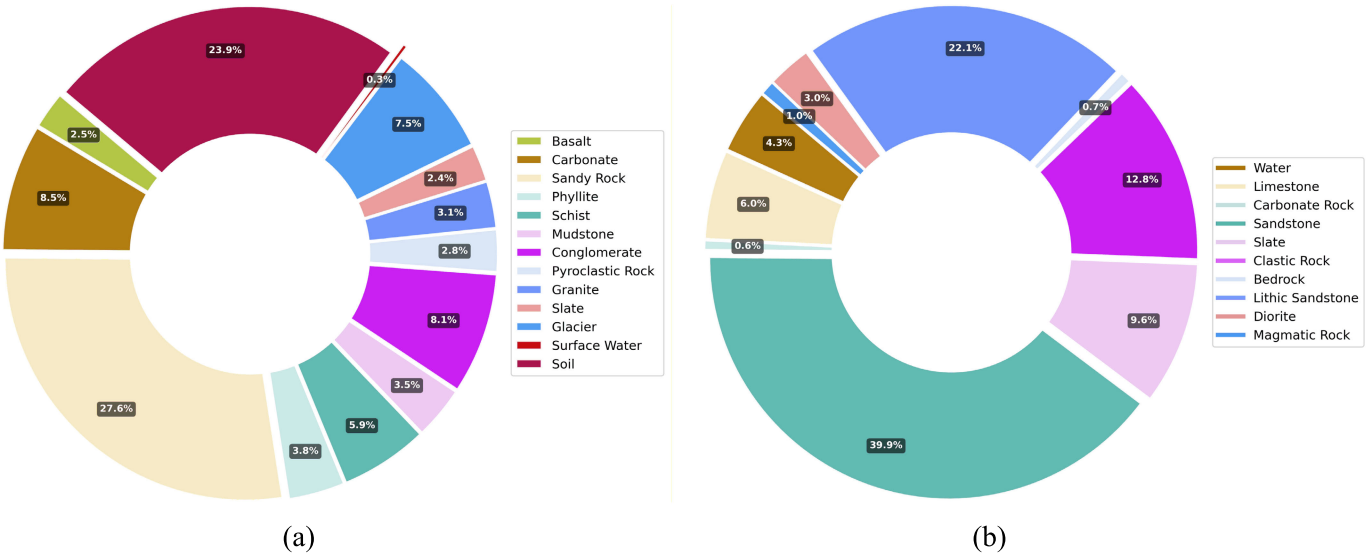


Fig. 4. Distribution of geological category proportions in (a) Geo-WQ and (b) Geo-XZ datasets.

Landsat-8 images were obtained from the Geospatial Data Cloud platform; bands 1–7 and 9 were selected to provide spectral information, while the 15 m panchromatic band (Band 8) was utilized for spatial enhancement. Before dataset construction, we performed Gram–Schmidt pan-sharpening within ArcGIS to fuse the 30 m multispectral bands with the 15-m panchromatic data. This preprocessing effectively increased the spatial resolution of the final 9-band imagery to 15 m.

2) *Sentinel-2:* The primary mission of the Sentinel-2 satellite is to provide high-resolution, multispectral Earth observation data via its multispectral imager. In this research, we selected four 10 m resolution spectral bands (Bands 2, 3, 4, and 8). The Level-2A product data were retrieved and processed on the Google Earth Engine (GEE) platform, where cloud and shadow masking, followed by a median composition, were performed to generate high-quality cloud-free composite data. After these GEE-based steps, we can get the four bands with 10-m resolution Sentinel-2 image.
3) *GF-5:* A distinctive characteristic of the GF-5 satellite hyperspectral data is its broad spectral range and high band density. The satellite provides up to 330 spectral bands, covering wavelengths from the visible–near infrared (VNIR) to the short-wave infrared region (SWIR). To reduce storage and computational demands, we select a subset of bands from 330 bands. Based on the reflection and absorption characteristics of minerals [26], [27], [28], the corresponding absorption ranges of geological lithology elements in the study area are found from the GF-5 image bands. Ten bands are chosen from the VNIR range (Bands 21, 25, 26, 29, 31, 36, 63, 76, 80, and 150) and ten from the SWIR range (Bands 151, 200, 211, 261, 268, 269, 271, 275, 280, and 318). After preprocessing, the resulting GF-5 image comprises 20 bands with a spatial resolution of 30 m.

All RS images from two geological datasets were resampled in ArcGIS with the nearest neighbor interpolation method, and the spatial resolution of images was uniformly defined as 15 m, ensuring that the size of each image matched the corresponding annotation map and achieving pixel-level correspondence with its label grid.

III. Methodology

To address the issue of spatial heterogeneity in multisource RS images and the problem of boundary ambiguity in segmentation maps, this study developed MGALNet based on a multibranch structure. The overall framework of MGALNet is shown in Fig. 5. This model integrates three submodules: MGFEFM, FRM, and SFEM.

We used the images from the Geo-WQ dataset as a case to demonstrate the data change in applying these modules. Two images are fed into MGALNet; the images have sizes of [9,224,224] and [20,224,224] for Landsat-8 and GF-5, respectively. These RS data are processed by convolutional operations to generate features with size [64,112,112], which are input to MGFEFM. Four MGFEFMs are connected in series during the feature extraction and cross-source fusion stage. Three feature vectors at different scales are output at this stage and referred to as low-, mid-, and high-level features, with vector sizes of [128,56,56], [256,28,28], and [1024,14,14], respectively. Then, through an adaptive average pooling operation, an aggregate feature that includes three-scale features is acquired, and the size is [1408,14,14]. Subsequently, the spatial context feature of elements is enhanced by FRM, whose size remains the same. The enhanced semantic feature vectors are separated and upsampled to match the original scales [128,56,56], [256,28,28], and [1024,14,14] at the feature-split stage. Given the complementarity of multiscale geological features and the spatial correlation and spectral continuity of geological elements in pixels, the multiscale spatial context feature fusion stage integrates semantic features at different scales and spatial features

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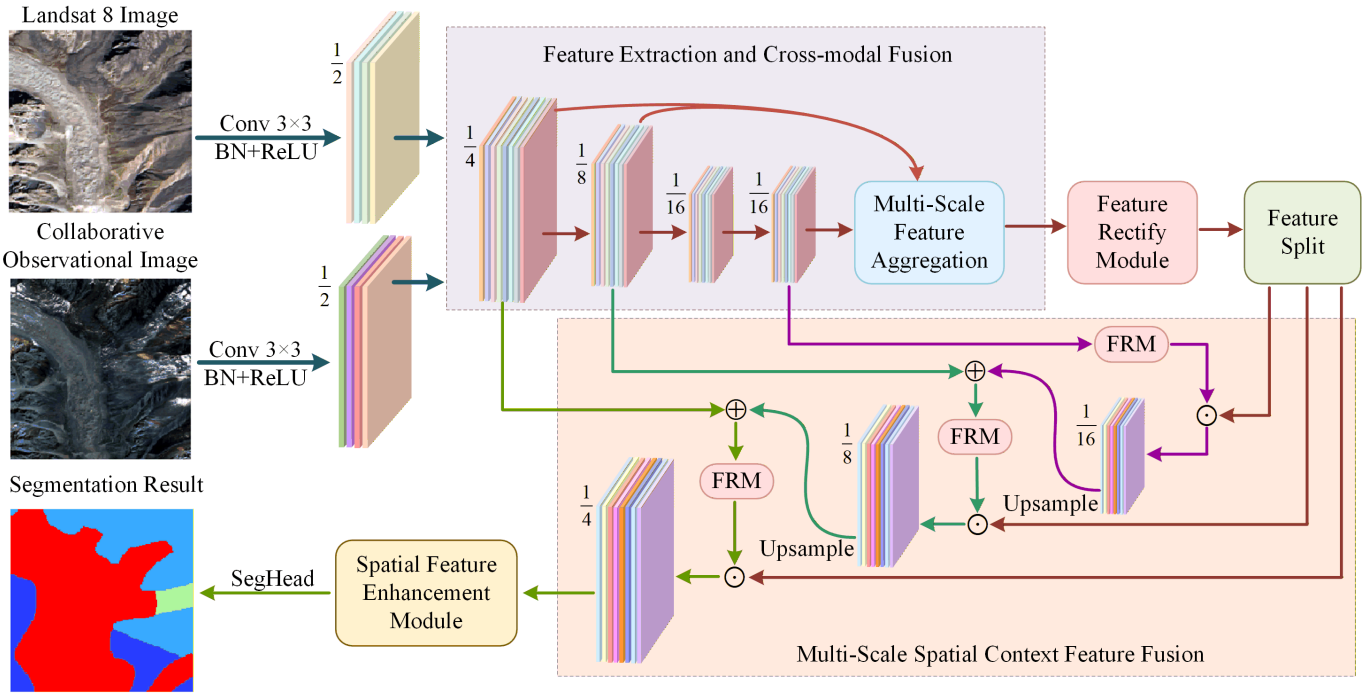


Fig. 5. Overall framework of the multigranularity feature adaptive network that consists of three submodules: the MGFEEM, the feature rectify module, and the SFEM.

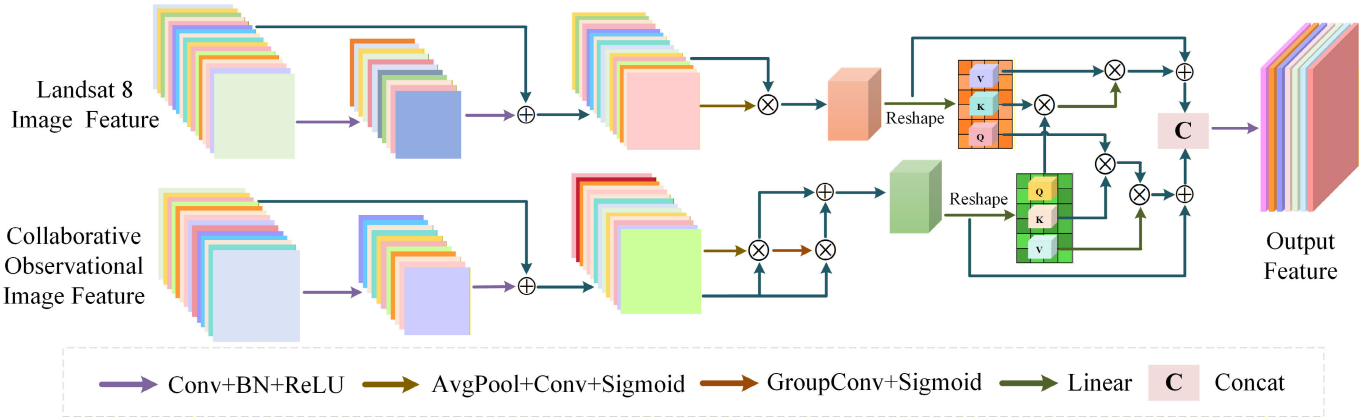


Fig. 6. Structure of the MGFEEM. This module mitigates spatial heterogeneity in multisource features by double-branch granularity extraction and adaptive feature fusion, ensuring consistent feature representation.

enhanced by FRM. This process effectively merges multiscale contextual information and enhances spatial features, producing a reconstructed feature map of size [128,112,112]. The feature vector is forwarded to SFEM for cross-spatial information aggregation, enhancing the spatial detail representation of adjacent elements. The enhanced feature of size [128,224,224] is sent to SegHead, generating its segmentation map.

A. Multigranularity Feature Extraction and Fusion Module

Variations in data acquisition principles and characterization dimensions cause the same geological element to exhibit inconsistent spatial distributions and representation intensities across RS sources. Consequently, element features exhibit substantial spatial heterogeneity in multisource geological images, leading to information conflicts when multisource features are directly fused, and making it difficult to leverage the collaborative advantages of multisource RS data. To address this issue, we propose MGFEFM that can effectively utilize the complementary information in multisource images and enhance the feature utilization rate of geological elements at the feature extraction and fusion stage. The structure of MGFEFM is illustrated in Fig. 6. MGFEFM has a dual-branch structure with two input data streams. The first branch captures the element feature in the Landsat-8 image. It processes the data through three layers of convolution operations to obtain a local coarse-grained feature. Subsequently, feature selection and weighting operations highlight important local feature information while suppressing less notable features. The second branch extracts features from the collaborative

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observational image by utilizing three layers of convolution to produce a local coarse-grained feature.

Before integrating these cross-model features from the two branches, we employ a fine-grained feature-gating mechanism to create a fine-grained feature-focusing area within the collaborative observation feature map. Specifically, the mask operation zeros spatial features with values below the threshold (0.001) to suppress noise and irrelevant information, emphasize key spatial regions, and enhance feature sparsity and discriminability. Subsequently, the GroupConv operation captures salient characteristics along the feature channel dimension, generates a spectral feature map, and applies adaptive weighting to the original data, thereby strengthening the representation of spectral features. The workflow of this process is given as

$$f_1 = \sigma\left(\text{Conv}\left(\text{AvgPool}\left(x\right)\right)\right) \tag{1}$$

$$f_2 = x \times mask\left(f_1\right) \tag{2}$$

$$f_3 = f_2 + f_1 \times \sigma\left(\text{GroupConv}\left(f_2\right)\right) \tag{3}$$

where x is the input data, f_1 represents the feature after the first processing operation, AvgPool$(\cdot)$ refers to the average pooling operation, Conv$(\cdot)$ denotes the ordinary convolution operation, $\sigma(\cdot)$ is the sigmoid activation function, and mask$(\cdot)$ is a mask function that is based on the threshold (0.001) and used for feature selection to suppress noise feature. f_2 refers to the features after refining spatial information, GroupConv$(\cdot)$ is the group convolution operation used in the channel dimension, and f_3 denotes the output features of the fine-grained feature gating mechanism.

Given that the coarse- and fine-grained features in RS images interact and complement each other, MGALNet should leverage the multigranularity feature information from various image sources. To achieve this goal, this module employs a cross-source feature fusion strategy to fuse the features of the two branches. This operation improves the feature utilization rate of geological elements and enhances the geological feature semantic perception ability of MGALNet. The workflow of this stage is shown as

$$f_{i1}, f_{i2} = \text{CProcess}\left(x_i\right) \tag{4}$$

$$q_i, k_i, v_i = \text{R}\left(f_{i2}\right), \text{CProcess}\left(f_{i2}\right) \tag{5}$$

$$F_1 = f_{11} + \text{L}\left(\text{S}\left(q_2 k_1^T / \sqrt{N_k}\right)\right) \times v_1 \tag{6}$$

$$F_2 = f_{21} + \text{L}\left(\text{S}\left(q_1 k_2^T / \sqrt{N_k}\right)\right) \times v_2 \tag{7}$$

$$F_{out} = \text{ConvBL}\left(\text{Cat}\left(F_1, F_2\right)\right). \tag{8}$$

Here, $i \in (1, 2)$ refers to the data sourced from different branches, and CProcess$(\cdot)$ is the channel processing function, which consists of the R$(\cdot)$ and the L$(\cdot)$ function. R$(\cdot)$ is the reshape operation, L$(\cdot)$ is the Linear operation, and S$(\cdot)$ is the softmax operation. This function can process input features x_i of the branch i into two feature vectors (f_{i1} and f_{i2}) with the same number of channels. q_i, k_i, v_i are the three feature vectors after processing features f_{i2}, and N_k is the channel number of k vector. F_1 represents the interaction feature of Branch 1, and F_2 is the interaction feature of Branch 2. Cat$(\cdot)$ refers to the channel concatenation operation. ConvBL$(\cdot)$ is the operation that combines the methods of ordinary convolution, normalization, and activation function. F_{out} represents the output feature after cross-source fusion.

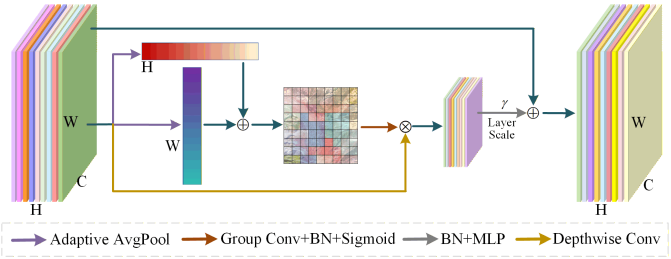


Fig. 7. Structure of the feature rectify module. This module integrates the spatial context in vertical and horizontal directions, enhancing spatial feature representation.

B. Feature Rectify Module

The edge regions of geological elements often present similar features, and a complex spatial structure with limited feature representation capability makes it difficult to segment the spatial boundaries precisely. Thus, the problem, like boundary ambiguity segmentation, often occurs in the segmentation maps. Inspired by the improvement in the position modeling of foreground objects [29], we propose FRM to achieve adaptive enhancement of spatial context information and enhance the scale-aware semantic feature ability of MGALNet. The structure of FRM is shown in Fig. 7. This module enables MGALNet to capture spatial context information in horizontal and vertical directions while minimizing the effect of irrelevant noise on edge features. FRM consists of three feature-processing branches. The first two branches employ adaptive average pooling operations to extract features from horizontal and vertical directions to emphasize horizontal continuity and enhance vertical structural correlation among features.

Therefore, it allows MGALNet to capture and utilize spatial features in 2-D effectively. After fusing these spatial context features from two directions, group convolution and the sigmoid function are applied to generate a regional attention weight map. In the third branch, deep convolution is used to extract local detail information in the original features. Then, the important regions of the feature are adaptively adjusted by dot with the attention weight map, producing a context-enhanced feature. In addition, the batch normalization operation and the multilayer perceptron based on 1×1 convolution with the layer-scale operation are jointly used to achieve dynamic adjustment in feature spectral information while preserving the spatial feature. Afterward, the attention weight map is dotted with the original features through the residual skip operation. The basic information in the original features is retained, and the importance of spatial context features is highlighted by applying the residual skip operation, which achieves the collaborative optimization of original and spatial context-enhanced features. The process of feature correction by this module is described as follows:

$$F_{HW} = \text{AdaAvgPool}_H\left(x\right) \oplus \text{AdaAvgPool}_W\left(x\right) \tag{9}$$

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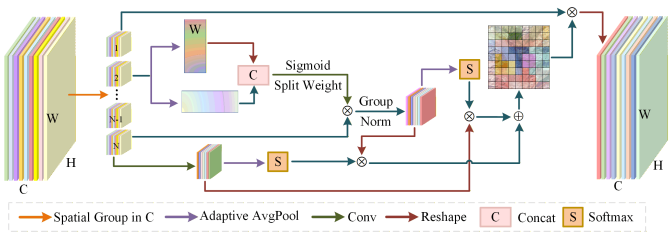


Fig. 8. Structure of the SFEM. This module can capture spatial dependencies across channel dimensions to further enhance boundary feature discrimination of adjacent elements.

$$f_{atten} = \sigma\left(\text{GroupConvBN}\left(F_{HW}\right)\right) \tag{10}$$

$$F_{out} = \gamma \cdot \text{BMLP}\left(\text{DepthConv}\left(x\right) \otimes f_{atten}\right) \oplus x \tag{11}$$

where $\text{AdaAvgPool}_{\text{d}}(\cdot)$ refers to the adaptive average pooling operation used in one direction, and $d \in (H, W)$ represents the horizontal and vertical directions, respectively. F_{HW} denotes the feature that fuses the context information from two directions. GroupConvBN$(\cdot)$ consists of two grouped convolution operations and batch normalization operations, $\sigma(\cdot)$ refers to the sigmoid function, and f_{atten} is the regional attention feature map. Moreover, DepthConv$(\cdot)$ refers to the deep convolution operation, and BMLP$(\cdot)$ includes a batch normalization operation and a multilayer perceptron based on 1×1 convolution. γ is a learnable channel scaling weight coefficient, its layer-scale operation assigns a dynamically adjusted weight to each feature channel to achieve adaptive correction of element features.

C. Spatial Feature Enhancement Module

In geological element semantic segmentation, accurate division of edge regions depends not only on local features to learn the shape and texture but also on the spatial relationship between geological elements. However, the CNN-based segmentation model is limited by the local receptive field in the convolution kernel, making it difficult to efficiently capture the long-distance dependencies between the features in the image. Inspired by cross-dimensional space learning [30], we design SFEM to enable MGALNet to maximize the use of global context information, and its structure is shown in Fig. 8. SFEM can adaptively adjust the importance of spectral features, retain the local spatial structure information of features, and capture the spatial dependencies between elements across channels, thus providing abundant spatial and spectral detail information for MGALNet training.

The number of input feature channels is divided into N group features, allowing MGALNet to learn rich spectral semantic information. SFEM consists of two feature-processing branches. The first branch adopts two parallel adaptive average pooling operations, which extract local spatial structure information from the horizontal and vertical spatial directions. It can obtain the attention weight map of grouped features through convolution operations and the sigmoid function, enhancing the important region of local spatial features. The second branch utilizes convolution and adaptive average pooling operations in series to further mine the scale-space information of the features. It can also obtain another attention weight map of the grouped features by applying the softmax function. The grouped features continuously and interactively fuse and aggregate with the attention weight maps in the two branches to produce a global spatial attention map. Then, weighted with the original features, spatial features enhanced with global context information are obtained. The process of SFEM can be described as follows:

$$f_{1i} = x_i \otimes \text{S}\left(\text{Conv}\left(\text{Concat}\left(\text{AdaAvgPool}_{H}\left(x_i\right) \times \text{AdaAvgPool}_{W}\left(x_i\right)\right)\right)\right) \tag{12}$$

$$f_{2i} = \text{Conv}\left(x_i\right) \tag{13}$$

$$A_i = \text{R}\left(\text{GN}\left(f_{1i}\right)\right) \otimes \text{APS}\left(f_{2i}\right) + \text{R}\left(f_{2i}\right) \otimes \text{APS}\left(f_{1i}\right) \tag{14}$$

$$f_{out} = \text{R}\left(\sum_{i=1}^{N} x_i \otimes A_i\right). \tag{15}$$

Here, x_i refers to the grouping feature, $x = [x_1, x_2, \ldots, x_N]$, $x_i \in x^{C/N \times H \times W}$, f_{1i} is the grouping feature output of Branch 1, f_{2i} refers to the grouping feature output of Branch 2, $\text{AdaAvgPool}_{\text{h,w}}(\cdot)$ is the adaptive average pooling operation in horizontal direction or vertical direction, Concat$(\cdot)$ refers to the channel splicing operation, Conv$(\cdot)$ denotes the convolution operation, and S$(\cdot)$ is the softmax function. A_i refers to the global attention weight map that aggregates the grouped features and weight with the local attention weight maps in Branches 1 and 2, GN$(\cdot)$ refers to the group normalization operation, APS$(\cdot)$ includes the adaptive average pooling operation and softmax function, R$(\cdot)$ refers to the reshape channel operation, and f_{out} refers to the output feature.

D. Binary Supervised Loss Function

To expedite model convergence and address the issue of class imbalance within the datasets, MGALNet should capture the crucial characteristics of geological elements in multisource RS data. Inspired by the hard example mining algorithm [22], we propose a binary supervised loss function to facilitate the training of MGALNet. The loss function is constructed using the fusion results of the mid-level feature from MGFEFM, the mid-level feature from the multiscale spatial context feature fusion stage, and the output feature of MGALNet to establish a deep supervision signal.

In addition to employing the cross-entropy loss function, the binary supervised loss function incorporates a hard sample mining loss function. This loss function determines the threshold for difficult samples on the basis of hyperparameters within the algorithm before calculating the cross-entropy loss. Specifically, the cross-entropy loss for each pixel in the sample is computed by (19). The loss values calculated from all samples N are then sorted in ascending order to obtain a set of loss values $L_{sort}\{loss_i\}_{i=1}^{N}$. The computation of hard sample loss can be expressed as

$$loss_i = -\sum_{c=1}^{C} y_{i,c} \log\left(L_{i,c}\right) \tag{16}$$

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where $y_{i,c}$ is the predicted label of the pixel, and $L_{i,c}$ represents the true label of the pixel

$$L_{hard} = \begin{cases} \{loss_i | loss_i > \tau\}, & \text{If } L_{sort}[N_{\min}] > \tau \\ \{L_{sort}[i]\}_{i=1}^{N_{\min}}, & \text{otherwise} \end{cases} \tag{17}$$

$$Loss_m = \frac{1}{|L_{hard}|} \sum_{loss_i \in L_{hard}} loss_i \tag{18}$$

where L_{hard} is the set of hard samples, τ represents the loss threshold, and N_{min} is the minimum number of samples involved in the loss calculation (set to 10 000 in this function).

To mitigate the excessive reliance of MGALNet on intermediate features, which affects the final segmentation result, we apply an additional hyperparameter weight factor to the cross-entropy loss term. The overall loss function can be expressed as

$$Loss_{all} = Loss_m(\tau, f_{out}, f_L) + \alpha \times Loss_c(f_{mid}, f_L) \tag{19}$$

where f_{out} is the output feature of MGALNet, f_{mid} is the mid-level feature of fusion, f_L represents the label data, $Loss_m$ is the hard sample mining function, $Loss_c$ denotes the cross-entropy loss function, and α is the weight factor in the binary supervised loss function.

IV. Experiments and Analysis

A. Experimental Settings

To reduce computational costs, we preprocessed the Landsat-8, GF-5, Sentinel 2, and labeled images into nonoverlapping patches, each with a size of 224 × 224 pixels. This operation resulted in 1140 unique patches from the Geo-WQ dataset, and 1280 nonoverlapping patches were extracted from the Geo-XZ dataset. The batch of these images was randomly divided into three sets: 80% for model training, 10% for model validation, and the remaining 10% for model testing. This study utilized seven deep learning-based multisource semantic segmentation models for comparative experiments; these models are DFAFNet [31], ASMFNet [32], FTransUNet [33], CMFNet [34], SFADFNet [3], CMNeXt [35], and GFBNeXt [36]. All experiments in this study were conducted on the NVIDIA L20 GPU workstation. The batch size was set to 18, and the initial learning rate of the models was set to 0.001. The epoch size was set to 200 for all multisource semantic segmentation models. To quantitatively evaluate the performance of different segmentation methods, we calculated four evaluation indicators by using a confusion matrix to analyze the results. The four indicators were class pixel accuracy (CPA), overall accuracy (OA), intersection over union (IoU), and mean IoU (mIoU). The overall evaluation indicators, OA and mIoU, were used to evaluate the segmentation result of each model. The calculation process of these indicators can be described as

$$CPA_i = \frac{P_{ii}}{\sum_{j=1}^{N} p_{ij}} \tag{20}$$

$$OA = \frac{\sum_{i=1}^{N} P_{ii}}{\sum_{i=1}^{N} \sum_{j=1}^{N} P_{ij}} \tag{21}$$

where N represents the number of geological element categories, P_{ii} refers to the number of pixels whose actual category is correctly predicted as category i, and P_{ij} is the number of pixels whose actual category should be predicted as category i but are predicted as category j

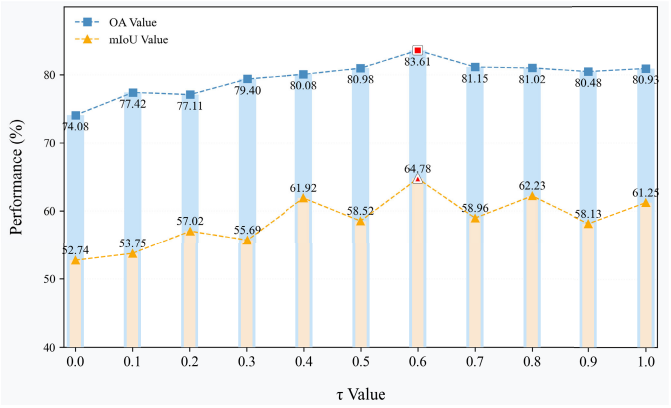


Fig. 9. Result of MGALNet for τ parameter in the Geo-WQ dataset.

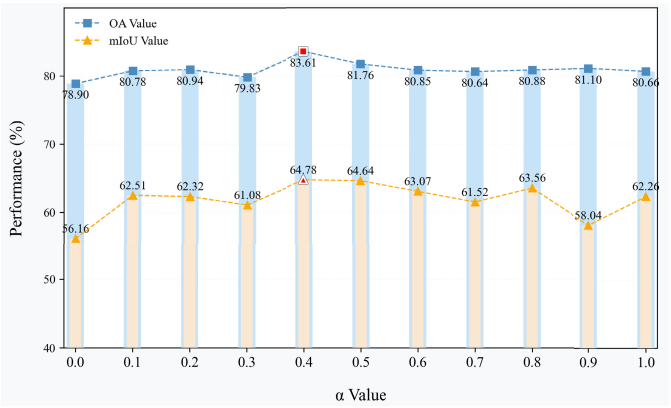


Fig. 10. Result of MGALNet for α parameter in the Geo-WQ dataset.

$$IoU = \frac{|A \cap B|}{|A \cup B|} \tag{22}$$

$$mIoU = \frac{\sum_{i=1}^{N} IoU_i}{N}. \tag{23}$$

Here, A refers to the number of actual values of a category, B denotes the number of predicted values of the category, N represents the number of geological element categories, and mIoU is the result of calculating the IoU values of all categories and taking the average.

To explore the optimal values of the two hyperparameters (i.e., hard sample recognition threshold τ and weight factor α) in the binary supervised loss function, we applied a numerical value of 0.1 within the range [0, 1] for each hyperparameter to complete a series of experiments in the Geo-WQ dataset. By evaluating the performance of MGALNet through OA and mIoU, we analyzed the different values of the hyperparameters that affected the segmentation results to identify the optimal values. Fig. 9 illustrates the segmentation results of MGALNet that uses various values for hard sample recognition threshold τ. and Fig. 10 presents the segmentation results for different values of weight factor α. The experimental results indicate that when the threshold τ is set to 0.6, and the weight factor

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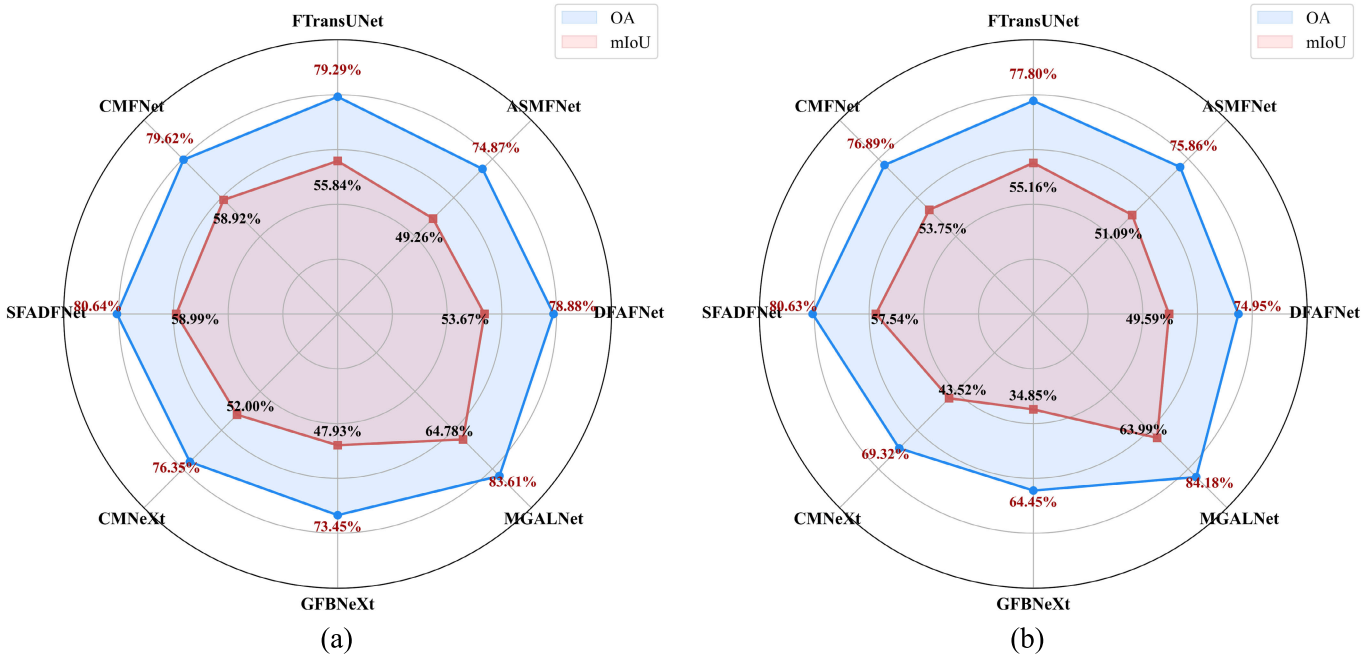


Fig. 11. OA and mIoU performance of the comparative models in datasets. (a) Performance in Geo-WQ. (b) Performance in Geo-XZ.

TABLE I
CPA AND OA VALUES OF DIFFERENT MODELS IN GEO-WQ DATASET

Class \ Method	DFAFNet [31]	ASMFNet [32]	FTransUNet [33]	CMFNet [34]	SFADFNet [3]	CMNeXt [35]	GFBNeXt [36]	MGALNet
Basalt	91.95	84.79	93.39	91.76	85.75	83.51	90.29	**98.57**
Carbonate	51.04	52.71	60.69	50.61	49.02	53.56	48.74	**68.99**
Sandy rock	80.18	77.43	80.02	78.60	**84.16**	76.99	70.61	82.41
Phyllite	67.77	63.16	72.63	**78.69**	69.03	70.85	75.26	76.49
Schist	62.34	59.68	65.30	63.11	72.64	54.77	**75.75**	72.51
Mudstone	34.45	30.84	47.70	46.27	**68.99**	41.00	38.46	67.63
Conglomerate	73.66	73.27	68.55	66.08	77.64	70.46	68.87	**82.44**
Pyroclastic rock	81.72	70.71	82.26	82.09	86.16	**95.03**	75.94	75.21
Granite	**79.97**	62.11	76.42	78.60	77.25	67.17	60.99	76.38
Slate	56.22	53.24	69.18	71.83	**83.86**	44.81	55.56	81.65
Glacier	88.43	**93.24**	88.70	88.97	85.77	91.08	90.28	89.09
Surface water	12.12	3.19	20.49	**46.81**	16.54	35.16	0.00	40.07
Soil	90.01	83.34	87.76	**91.89**	88.21	86.24	80.77	91.56
OA	78.88	74.87	79.29	79.62	80.64	76.35	73.45	**83.61**

α is set to 0.4, MGALNet produces an excellent segmentation result.

B. Experimental Results and Analysis

Fig. 11(a) presents the OA and mIoU performance of the compared models. Table I lists the CPA and OA values for the different segmentation models, and Table II displays the IoU and mIoU values.

Meanwhile, Fig. 12 presents the segmentation results of the various models applied to the Geo-WQ geological dataset, and Fig. 13 shows the segmentation results of each model in the local region of Geo-WQ. The experimental results in Tables I and II reveal that among the seven compared methods, GFBNeXt exhibits the worst segmentation performance, followed closely by ASMFNet. By contrast, the proposed MGALNet achieves the best segmentation results, with an impressive OA of 83.61% and an average mIoU of 64.78%. Compared with ASMFNet and GFBNeXt, MGALNet shows OA increments of 8.74% and 10.16%, respectively, and mIoU increments of 15.52% and 16.85%, respectively. The segmentation results in Fig. 13(c) and (h) further reveal that the last two models contain more error regions compared with MGALNet. In comparison with the classic multisource semantic segmentation model CMNeXt, which is based on an encoder–decoder architecture, MGALNet demonstrates increments of 7.26% in OA and 12.78% in mIoU. Moreover, compared with the OA value of DFAFNet, the OA value of MGALNet improves by 4.73%, and its mIoU increases by 11.11%. Fig. 13(b), (i), and (g) also shows that MGALNet results in only a few incorrect segmentations in the intersecting edge areas of geological elements. These results demonstrate the superiority of MGALNet, which effectively captures multigranularity features and the local–global scale spatial information of geological elements. Compared with FTransUNet, MGALNet has a 4.32% higher OA value and 8.94% higher mIoU. The segmentation results in Fig. 13(d) and (i)

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TABLE II
IoU AND mIoU VALUES OF DIFFERENT MODELS IN GEO-WQ DATASET

Class \ Method	DFAFNet [31]	ASMFNet [32]	FTransUNet [33]	CMFNet [34]	SFADFNet [3]	CMNeXt [35]	GFBNeXt [36]	MGALNet
Basalt	75.39	74.06	67.30	77.13	77.93	66.62	**79.07**	78.56
Carbonate	46.05	42.51	50.78	44.16	42.99	41.72	42.34	**55.96**
Sandy rock	68.43	61.25	68.28	67.21	69.28	64.35	58.63	**73.72**
Phyllite	49.44	53.64	58.78	60.86	59.93	60.68	49.83	**66.37**
Schist	43.82	41.73	53.55	52.07	51.37	42.54	48.13	**56.85**
Mudstone	27.38	27.05	34.74	30.96	**49.88**	30.89	20.34	47.06
Conglomerate	67.24	53.58	57.16	56.73	71.52	54.08	41.84	**77.62**
Pyroclastic rock	47.09	46.70	41.50	**63.11**	46.90	45.07	38.82	61.63
Granite	**61.82**	50.78	58.46	60.13	61.52	50.01	50.37	61.35
Slate	47.08	44.16	53.61	58.27	64.69	39.50	38.03	**64.72**
Glacier	77.76	72.04	77.33	76.86	78.12	76.00	77.02	**79.12**
Surface water	10.79	2.91	27.22	**42.33**	15.22	30.24	0.00	39.44
Soil	75.43	70.10	77.23	76.21	77.64	74.35	70.21	**78.92**
mIoU	53.67	49.26	55.84	58.92	58.99	52.00	47.93	**64.78**

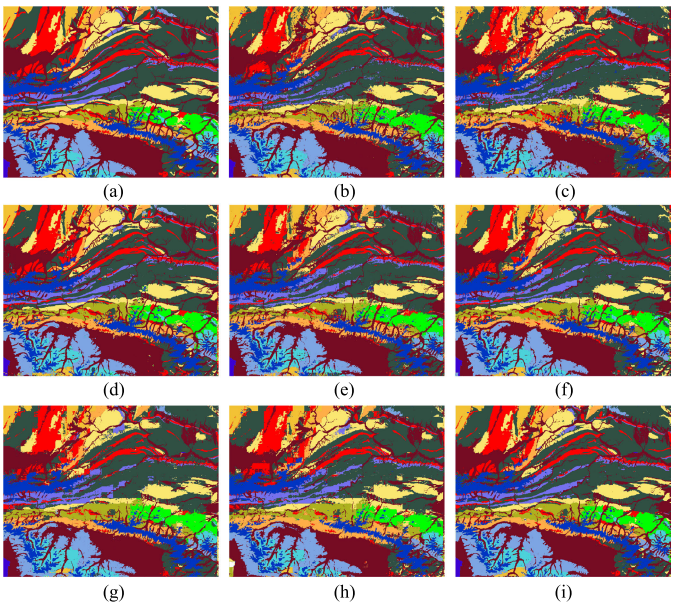


Fig. 12. Segmentation results of comparative models on Geo-WQ dataset. (a) Ground truth. (b) DFAFNet. (c) ASMFNet. (d) FTransUNet. (e) CMFNet. (f) SFADFNet. (g) CMNeXT. (h) GFBNeXt. (i) MGALNet.

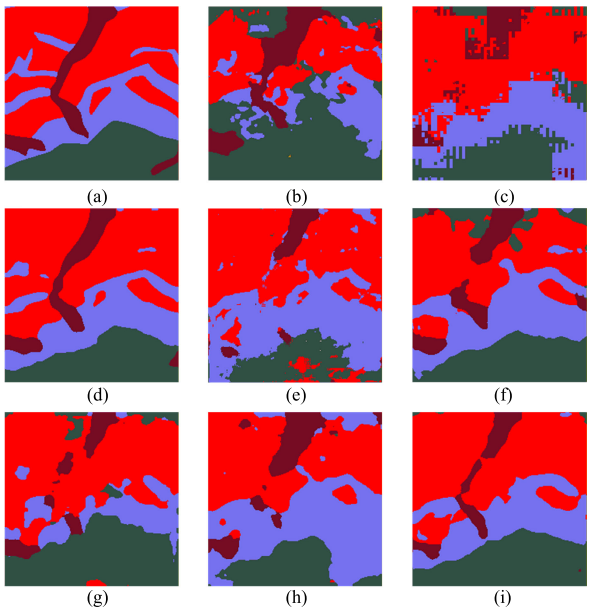


Fig. 13. Predicted example of comparative models on Geo-WQ dataset. (a) Ground truth. (b) DFAFNet. (c) ASMFNet. (d) FTransUNet. (e) CMFNet. (f) SFADFNet. (g) CMNeXT. (h) GFBNeXt. (i) MGALNet.

confirm that MGALNet outperforms FTransUNet in terms of segmentation results. Meanwhile, compared with CMFNet, which is based on CNN and Transformer, MGALNet shows improvements of 3.99% in OA and 5.86% in mIoU, indicating that it enhances segmentation accuracy while reducing computational complexity. This conclusion is also supported by the results in Fig. 13(e) and (i). In addition, compared with the latest multisource semantic segmentation model SFADFNet, MGALNet shows an improvement of 2.97% in OA and 5.79% in mIoU. Fig. 13(i) and (f) also reveals that SFADFNet has more missegmented areas of geological elements than MGALNet, reinforcing the overall superiority of MGALNet. Notably, several models have low segmentation accuracy for Category 12 (surface water) in Tables I and II. The primary reason for this is the unbalanced distribution of geological element samples in the dataset. Specifically, the proportion of surface water samples is less than 0.3% of the total samples in the Geo-WQ dataset. Splitting the images directly can lead to an uneven distribution of surface water in the training images, which occupy a small portion of all the pixels. As a result, the segmentation model tends to focus on learning the features of other types and fails to learn surface water characteristics effectively. Despite this issue, MGALNet still demonstrates superior segmentation performance for surface water, affirming its effectiveness.

Fig. 11(b) presents the performance of the compared models in Geo-XZ. Table III lists the CPA and OA values for the different segmentation models, and Table IV shows the IoU and mIoU values for each model. Meanwhile, Fig. 14 illustrates the segmentation results of the various models applied to the Geo-XZ dataset, and Fig. 15 provides the segmentation results of each model in the local region of Geo-XZ. Notably, MGALNet achieves an impressive OA of 84.18% and an average mIoU of 63.99% in the experimental results. Fig. 11(b) indicates that MGALNet outperforms the other segmentation models. Among all the methods, GFBNeXt

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TABLE III
CPA AND OA VALUES OF DIFFERENT MODELS IN GEO-XZ DATASET

Class \ Method	DFAFNet [31]	ASMFNet [32]	FTransUNet [33]	CMFNet [34]	SFADFNet [3]	CMNeXt [35]	GFBNeXt [36]	MGALNet
Water	58.61	**69.46**	68.30	64.94	60.04	44.41	39.85	69.29
Limestone	73.70	79.90	81.25	77.89	**81.43**	57.44	49.77	81.02
Carbonate rock	43.44	66.69	66.09	75.23	73.23	61.45	0.00	**77.66**
Sandstone	82.94	79.42	83.79	76.42	**88.17**	77.02	76.35	87.85
Slate	48.71	74.88	61.57	75.90	71.00	52.84	61.09	**80.20**
Clastic rock	52.61	54.39	60.48	67.32	60.32	66.59	38.68	**69.86**
Bedrock	52.08	72.53	65.47	70.07	51.90	28.08	51.63	**83.91**
Lithic sandstone	88.19	82.32	85.00	86.59	87.62	74.13	74.59	**91.71**
Diorite	**93.09**	82.77	81.08	81.72	81.68	82.27	45.43	91.73
Magmatic rock	33.40	21.59	34.70	32.15	24.80	27.94	18.88	**36.36**
OA	74.95	75.86	77.80	76.89	80.63	69.32	64.45	**84.18**

TABLE IV
IOU AND MIOU VALUES OF DIFFERENT MODELS IN GEO-XZ DATASET

Class \ Method	DFAFNet [31]	ASMFNet [32]	FTransUNet [33]	CMFNet [34]	SFADFNet [3]	CMNeXt [35]	GFBNeXt [36]	MGALNet
Water	46.46	41.70	49.53	49.90	45.46	39.04	31.74	**55.69**
Limestone	61.31	65.13	62.25	64.60	69.97	50.17	39.37	**72.02**
Carbonate rock	37.75	49.74	50.68	50.25	**65.89**	39.40	0.00	64.61
Sandstone	65.93	67.30	70.77	66.57	73.67	60.73	56.53	**77.73**
Slate	40.29	51.86	48.56	57.65	56.58	35.71	38.12	**68.53**
Clastic rock	43.92	45.16	47.81	49.46	50.19	43.33	26.92	**58.06**
Bedrock	25.76	34.10	**54.36**	30.56	42.26	21.19	43.98	47.42
Lithic sandstone	68.54	69.03	71.44	69.60	75.44	58.29	56.48	**78.49**
Diorite	81.34	68.06	71.39	75.76	72.75	65.35	43.88	**84.18**
Magmatic rock	24.63	18.91	24.89	23.18	23.04	22.06	11.55	**33.18**
mIoU	49.59	51.09	55.16	53.75	57.54	43.52	34.85	**63.99**

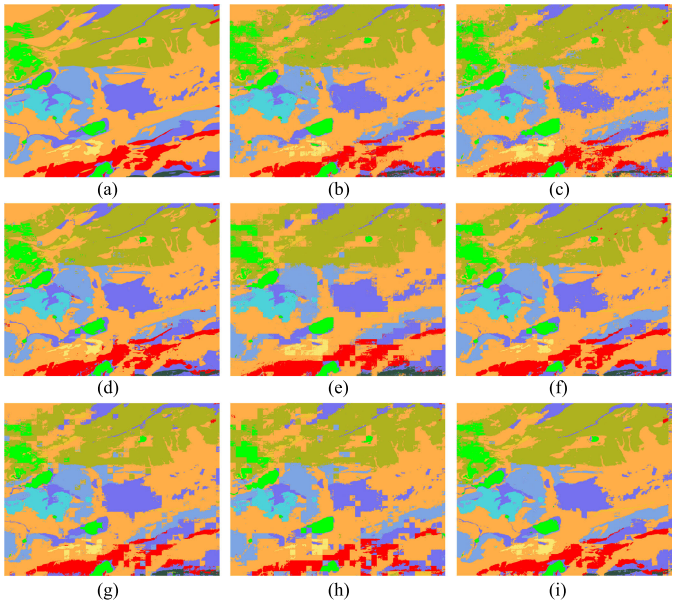


Fig. 14. Segmentation results of comparative models on the Geo-XZ dataset. (a) Ground truth. (b) DFAFNet. (c) ASMFNet. (d) FTransUNet. (e) CMFNet. (f) SFADFNet. (g) CMNeXT. (h) GFBNeXt. (i) MGALNet.

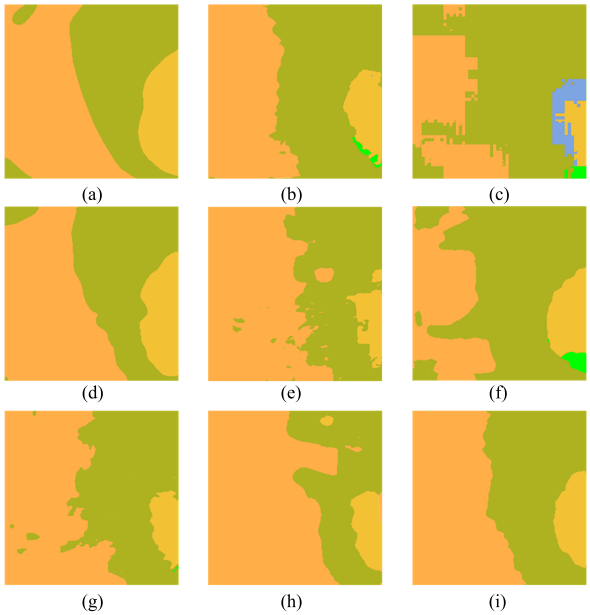


Fig. 15. Predicted example of comparative models on the Geo-XZ dataset. (a) Ground truth. (b) DFAFNet. (c) ASMFNet. (d) FTransUNet. (e) CMFNet. (f) SFADFNet. (g) CMNeXT. (h) GFBNeXt. (i) MGALNet.

exhibits the worst performance, followed by CMNeXt. Compared with the representative encoder–decoder multisource semantic segmentation method CMNeXt, MGALNet shows a substantial improvement, with OA and mIoU increments of 14.86% and 20.47%, respectively. The segmentation results in Fig. 15(g) show the presence of several misclassified regions in the results of CMNeXt. Meanwhile, MGALNet demonstrates better performance than CMFNet, which is based on CNN and Transformer, with increments of 7.29% in OA and 10.24% in mIoU. Fig. 15(i) shows that MGALNet has fewer misclassified geological elements than CMFNet. These results confirm that MGALNet effectively utilizes multigranularity features and local–global spatial information, leading to a reduction in model complexity and enhanced segmentation

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TABLE V

ABLATION RESULTS OF MGALNET AND SFADFNET IN GEO-WQ AND GEO-XZ DATASETS. AMONG THEM, EXPERIMENTS 1–4 ARE ABLATION EXPERIMENTS ON THE MGALNET, AND EXPERIMENTS 5–8 PRESENT THE RESULTS OF APPLYING EACH SUBMODULE TO THE SFADFNET MODEL, RESPECTIVELY

Experiment \ Method	SFADFNet	MGFEFM	FRM	SFEM	Geo-WQ		Geo-XZ	
					OA	mIoU	OA	mIoU
1	×	×	✓	✓	79.22	60.39	82.08	55.37
2	×	✓	×	✓	79.11	55.19	76.79	45.81
3	×	✓	✓	×	80.66	61.49	81.05	51.26
4	×	✓	✓	✓	**83.61**	**64.78**	**84.18**	**63.99**
5	✓	×	×	×	80.64	58.99	80.63	57.54
6	✓	✓	×	×	81.02	59.67	80.93	57.92
7	✓	×	✓	×	82.75	62.89	82.94	61.83
8	✓	×	×	✓	82.39	62.05	81.65	59.44

TABLE VI

ABLATION EXPERIMENTS OF MGALNET WERE CONDUCTED USING SINGLE-SOURCE IMAGES ON THE GEO-WQ AND GEO-XZ DATASETS

Experiment \ Data-source	Landsat-8 in Geo-WQ	Landsat-8 in Geo-XZ	GF-5	Sentinel-2	OA	mIoU
1	✓	×	×	×	78.28	54.51
2	×	✓	×	×	79.98	45.78
3	×	×	✓	×	72.69	45.91
4	×	×	×	✓	80.34	47.07

accuracy. Moreover, compared with the latest multisource semantic segmentation model SFADFNet, MGALNet achieves an OA increase of 3.55% and an mIoU increase of 6.45%. Fig. 15(i) and (f) further reveals that SFADFNet contains more misclassified geological element regions than MGALNet, reinforcing the superiority of MGALNet. In addition, Tables III and IV show that the segmentation accuracy of GFBNeXt for Category 3 (carbonate rocks) is 0 primarily because the proportion of carbonate rock samples in the Geo-XZ dataset is below 0.6%. Thus, the model fails to learn meaningful features from this category because of the limited pixel representation. By contrast, MGALNet displays much better performance for this category compared with most of the other models, underscoring the effectiveness of its structure.

V. ABLATION STUDY

This section presents a series of ablation experiments conducted on MGALNet, offering a comprehensive analysis and validation of each submodule. Due to space limitations, this section focuses on the experimental results obtained from the Geo-WQ dataset. In addition, we apply each submodule to the representative method, SFADFNet, to further demonstrate its effectiveness.

Table V displays the OA and mIoU values of the submodule in MGALNet and the performance of each submodule applied to SFADFNet in Geo-WQ and Geo-XZ datasets. Comparison of the results of Experiments 1 and 4 revealed a 4.39% increase in OA and mIoU values. This outcome effectively demonstrates the importance of the MGFEFM, highlighting the critical role that multigranularity feature information from various RS image sources plays in the interpretation ability of MGALNet. The effectiveness of the MGFEFM is further demonstrated by the results of Experiments 5 and 6, where SFADFNet achieved OA and mIoU improvements of 0.39% and 0.68%, respectively. Similarly, the comparison between Experiments 2 and 4 indicates that MGALNet achieves a 4.5% improvement in OA and a remarkable 9.59% gain in mIoU. This result validates the effectiveness of FRM and demonstrates the importance of the spatial context feature in enhancing the segmentation performance of MGALNet. The FRM enhances the spatial semantic representation capability of the model, as confirmed by Experiments 5 and 7, in which SFADFNet improves OA and mIoU by 2.11% and 3.9%, respectively. The effectiveness of SFEM is confirmed by comparing the results of Experiments 3 and 4, which show that the former outperforms the latter. The OA and mIoU improvements of 2.95% and 3.29% achieved by MGALNet further demonstrate the effectiveness of the SFEM in the lithology segmentation process. The results of Experiments 5 and 8 further demonstrate that this module enables the SFADFNet model to effectively learn the spatial dependencies among lithologies in RS images, improving the segmentation accuracy for each geological lithology. In summary, the design of each submodule in MGALNet is logical and effective, with each submodule playing a vital role in enhancing the overall performance of MGALNet.

As shown in Table VI, the performance of MGALNet is evaluated using single-source images from Geo-WQ and Geo-XZ datasets. The results demonstrate that MGALNet maintains stable segmentation accuracy under a single-source input, exhibiting a decline compared with multisource fusion. These results indicate that the proposed framework possesses a strong feature extraction capability and robustness. Compared with the results of Experiment 4 in Table V, the results

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TABLE VII
QUANTITATIVE COMPARISON OF EACH MODEL COMPLEXITY IN TERMS OF PARAMS AND FLOPS

Methods	Params (M)	FLOPs (G)
DFAFNet	12.45M	15.45G
ASMFNet	83.51M	49.14G
FTransUNet	179.42M	43.00G
CMFNet	98.51M	60.90G
SFADFNet	204.33M	208.43G
CMNeXt	58.73M	11.35G
GFBNeXt	9.38M	2.78G
MGALNet	58.96M	25.77G

of MGALNet achieve higher values in the OA and mIoU, confirming that complementary information across different RS sources effectively enhances lithology discrimination and boundary segmentation of MGALNet.

VI. EFFICIENCY ANALYSIS

Computational efficiency is a critical consideration in lithological interpretation tasks. To evaluate the computational efficiency of the proposed MGALNet, we compared the model complexity of different multisource segmentation methods in terms of the number of parameters (Params) and FLOPs, as summarized in Table VII. The results show that MGALNet achieves a favorable tradeoff between segmentation accuracy and computational cost. Although the parameter counts of MGALNet (58.96 M) are slightly higher than those of lightweight models, such as GFBNeXt (9.38 M), they remain substantially lower than those of large-scale models with good segmentation performance, such as SFADFNet (204.33 M) and FTransUNet (179.42 M). Furthermore, the FLOPs of MGALNet (25.77 G) are markedly reduced compared with those of SFADFNet (208.43 G), showing a nearly eightfold reduction in computational overhead. These results demonstrate that MGALNet achieves high segmentation accuracy while maintaining moderate model complexity and efficient inference, effectively balancing the performance and computational cost in multisource lithological interpretation tasks.

VII. CONCLUSION

This study proposes MGALNet, a multiple geological lithology interpretation framework designed to achieve adaptive multisource RS data fusion, reduce boundary ambiguity segmentation errors, and improve interpretation accuracy. MGFEFM enables adaptive cross-source feature fusion to address spatial heterogeneity in multisource RS data. FRM refines spatial structural features by integrating contextual features from horizontal and vertical directions, and SFEM models spatial dependencies across channels to achieve cross-dimensional spatial structure learning. Joint utilization of FRM and SFEM can mitigate the issue of boundary ambiguity segmentation. Evaluations on two newly constructed datasets, Geo-WQ and Geo-XZ, verify the superiority of MGALNet over existing methods. The results also demonstrate the robustness and effectiveness of MGALNet in utilizing different RS images across complex terrains for accurate geological lithological segmentation. However, multisource geological lithological interpretation still faces challenges, including sample imbalance in the dataset and degradation of model performance during the unified training of multiregion, multiclass lithological samples. Future research should design lightweight and efficient lithological interpretation frameworks and further explore their potential applications in unified training tasks with multiregion, multiclass lithological samples.

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